DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024 (Course: B.Tech.)

Branch : Electronics and Telecomm. Engg./ Electronics and Comm. Engg./ Electronics Engg.

Subject Code & Name: BTHM705 Engineering Economics and Financial Mathematics Semester : VII

Max Marks: 60 **Date:** 21/02/2025 **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

		(CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**		**12**
1	Firms ___ goods and services in the economy. a) demand b) supply c) restrict d) consume	CO1	1
2	The law of demand states that as the price of a good increases, the quantity demanded ___. a) increases b) decreases c) remains constant d) fluctuates	CO1	1
3	The equilibrium price is determined at the point where supply and demand ____. a) diverge b) intersect c) decrease d) stabilize	CO1	1
4	Sunk costs are expenses that cannot be ___. a) calculated b) recovered c) reduced d) controlled	CO1	1
5	The ratio of function to cost in value engineering is called: a) Performance b) Value c) Efficiency d) Reliability	CO2	1
6	A company decides to buy instead of making when it lacks _______. a) Space b) Equipment c) Demand d) Experience	CO2	1
7	The sinking fund factor helps in _______. a) Saving b) Spending c) Lending d) Borrowing	CO2	1
8	A revenue-dominated cash flow focuses on maximizing _______. a) Savings b) Costs c) Returns d) Taxes	CO2	1
9	A project is accepted if its Present Worth is _______. a) Zero b) Negative c) Positive d) Constant	CO2	1
10	Which maintenance type is performed at regular intervals to prevent failure? a) Corrective b) Predictive c) Preventive d) Breakdown	CO2	1
11	Which maintenance approach allows assets to run until they fail completely? a) Preventive b) Breakdown c) Condition-based d) Predictive	CO3	1
12	Which method applies an equal amount of depreciation each year? a) Declining b) Straight c) Sinking d) Sum	CO3	1

Q. 2	**Solve the following.**		**12**
A)	Explain the circular flow model of an economy with suitable examples.	CO1	**6**
B)	Describe the law of demand and supply with a suitable example.	CO1	**6**
Q.3	**Solve the following.**		**12**
A)	Define the make-or-buy decision. Describe criteria for make and buy.	CO2	**6**
B)	Ramesh wants to save money for his daughter's college education, which will cost Rs. 5,00,000 after 10 years. He has access to a fixed deposit scheme offering 6% annual interest. How much should Ramesh deposit today to ensure he has Rs. 5,00,000 after 10 years?	CO2	**6**
Q. 4	**Solve Any Two of the following.**		**12**
A)	Define cash flow. Describe types of cash flow with examples of each.	CO2	**6**
B)	Describe the concept of the Present Worth (PW) Method with example.	CO2	**6**
C)	A renewable energy project has: • Initial investment: Rs 5 crore • Annual income: Rs 1.5 crore for 5 years • Salvage value at the end: Rs 1 crore • Discount rate: 10% Calculate its present worth.	CO3	**6**
Q.5	**Solve Any Two of the following.**		**12**
A)	Discuss different types of maintenance with examples.	CO3	**6**
B)	Define economic life of an asset. What factors affect its determination?	CO3	**6**
C)	What do you mean by capital recovery? Describe the concept of Challenger vs. Defender.	CO3	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	What is depreciation? Explain its significance in financial planning and asset management.	CO3	**6**
B)	Describe the Straight-Line Method (SLM) of depreciation with an example.	CO3	**6**
C)	A textile factory purchases a machine for Rs. 10,00,000. The depreciation rate is 20% per year. Calculate the book value of the machine at the end of 3 years using the Declining Balance Method.	CO4	**6**

***** End *****